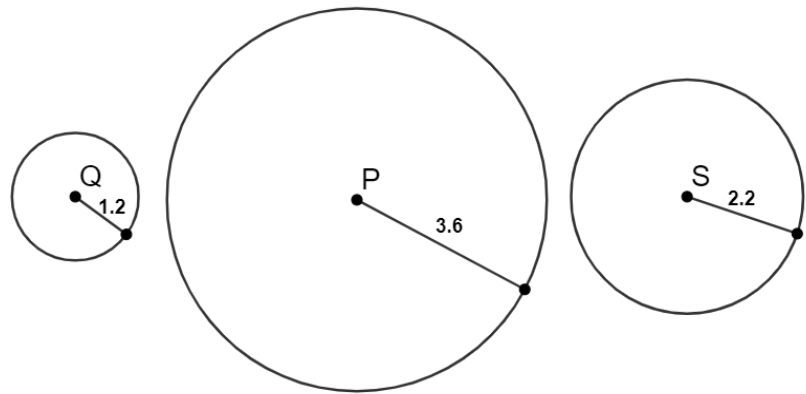


Circles Q , P , and S are shown. The radius of Circle Q is 1.2 units long. The radius of Circle P is 3.6 units long. The radius of Circle S is 2.2 units long.



1. Complete the following table. Round to the nearest thousandth. Use the π button on your calculator.

Circle	Radius	Circumference
Q	1.2	7.540
P	3.6	22.619
S	2.2	13.823

2. Complete the table.

Circle Q and P	
Ratio of Radius of Circle Q and Radius of Circle P	$\frac{1}{3}$
Ratio of Circumference of Circle Q and Circumference of Circle P	$\frac{1}{3}$

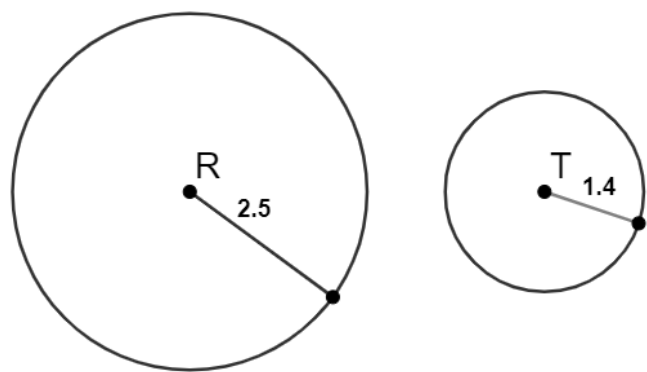
3. Complete the table.

Circle P and S	
Ratio of Radius of Circle P to Radius of Circle S	1.64
Ratio of Circumference of Circle P to Circumference of Circle S	1.64

4. Complete the table.

Circle Q and S	
Ratio of Radius of Circle Q to Radius of Circle S	0.545
Ratio of Circumference of Circle Q to Circumference of Circle S	0.545

Circles R and T are shown. The radius of Circle R is 2.5 units long. The radius of Circle T is 1.4 units long.



Show Circle R is similar to Circle T by completing the proof by contradiction:

5. Assume that Circle R is not similar to Circle T .
6. The ratio of the radius of Circle R to the radius of Circle T is 1.79.
7. The ratio of the circumference of Circle R to the circumference of Circle T is 1.79.
8. The ratio of the radius and the ratio of the circumference are equal, so corresponding parts of Circle R and Circle T are proportional.
9. Since the corresponding parts of Circle R and Circle T are proportional, the circles are similar. This is a contradiction, so the assumption that Circle R is not similar to Circle T is not true.
10. All circles are

o congruent

✗ similar

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